EXERCICE 1

We define the sequence u by $u_0 = 3$ and, for all natural numbers n, by the recurrence relation :

$$u_{n+1} = \frac{4u_n - 4}{u_n}.$$

1. Compute u_1, u_2, and u_3. Provide the values in irreducible fraction form.

2. For all natural numbers n, let

$$v_n = \frac{1}{u_n - 2}.$$

a) Compute the first four terms of the sequence v and conjecture its monotonicity.

b) Prove this conjecture by studying the difference $v_{n+1} - v_n$.